

ANSWER KEY — Integration Lesson 37 Test: Differential Equations I

For the grader · full working shown · give credit for correct method with small arithmetic slips · dock a point for two separate constants or a missing $\pm$ choice

1. Verify that $y = x^2 + 7$ is a solution of $dy/dx = 2x$.

Work: left side: $(x^2 + 7)' = 2x$. Right side: $2x$. The sides are equal for every x . **Answer: verified — both sides equal $2x$, so $y = x^2 + 7$ is a solution.**

2. Verify that $y = e^{(5x)}$ is a solution of $dy/dx = 5y$.

Work: left side: $(e^{(5x)})' = 5e^{(5x)}$ (chain rule). Right side: $5y = 5 \cdot e^{(5x)} = 5e^{(5x)}$. Equal for all x . **Answer: verified — both sides equal $5e^{(5x)}$.**

3. Show that $y = x^2$ is NOT a solution of $dy/dx = 2y$.

Work: left side: $(x^2)' = 2x$. Right side: $2y = 2x^2$. Is $2x = 2x^2$ for all x ? Test $x = 2$: left = 4, right = 8 — unequal. (Agreement at the special points $x = 0$ and $x = 1$ is not enough; a solution must work at EVERY x .) **Answer: not a solution — $2x \neq 2x^2$ in general (e.g. at $x = 2$).**

4. Solve $dy/dx = 3y$ (general solution).

Work: separate: $dy/y = 3 dx$. Integrate: $\ln|y| = 3x + C$. Exponentiate: $|y| = e^C \cdot e^{(3x)}$; absorb $\pm e^C$ into a single constant C . Check: $(Ce^{(3x)})' = 3Ce^{(3x)} = 3y$ ✓. **Answer: $y = Ce^{(3x)}$.**

5. Solve $dy/dx = 3x^2/y$ in the form $y^2 = \dots$

Work: separate: $y dy = 3x^2 dx$. Integrate: $y^2/2 = x^3 + C$. Multiply by 2 (C eats the 2): $y^2 = 2x^3 + C$. Check implicitly: $2y \cdot dy/dx = 6x^2 \rightarrow dy/dx = 3x^2/y$ ✓. **Answer: $y^2 = 2x^3 + C$.** (Accept $y^2/2 = x^3 + C$; dock for two constants.)

6. Solve $dy/dx = x/y$ with $y(0) = 6$.

Work: $y dy = x dx \rightarrow y^2/2 = x^2/2 + C \rightarrow y^2 = x^2 + C$. Plug in $(0, 6)$: $36 = 0 + C \rightarrow y^2 = x^2 + 36$. Then $y = \pm\sqrt{(x^2 + 36)}$; since $y(0) = 6 > 0$, take the $+$ root. Check: $dy/dx = x/\sqrt{(x^2 + 36)} = x/y$ ✓, $y(0) = \sqrt{36} = 6$ ✓; defined for all x . **Answer: $y = \sqrt{(x^2 + 36)}$, the positive root because $y(0) = 6 > 0$.**

7. Solve $dy/dx = 4xy$ with $y(0) = 2$.

Work: separate: $dy/y = 4x dx$. Integrate: $\ln|y| = 2x^2 + C$. Exponentiate: $y = Ce^{(2x^2)}$. Initial condition: $2 = C \cdot e^0 = C$. Check: $(2e^{(2x^2)})' = 2e^{(2x^2)} \cdot 4x = 4x \cdot y$ ✓, $y(0) = 2$ ✓. **Answer: $y = 2e^{(2x^2)}$.**

8. Columns identical, flat segments exactly on $x = 0$ — which DE fits?

Work: identical columns means moving up/down changes nothing, so the slope depends on x ONLY — the DE has the form $dy/dx = f(x)$. That rules out $dy/dx = y$ and $dy/dx = xy$ (both change along a column; and xy is zero on BOTH axes, not just $x = 0$). $dy/dx = x$ depends on x alone and is zero exactly when $x = 0$. **Answer: $dy/dx = x$ — slope depends on x only, zero precisely on the line $x = 0$.**

9. For $dy/dx = y(x - 1)$: where is the field flat, and what is the equilibrium solution?

Work: $y(x - 1) = 0$ when $y = 0$ (the whole x -axis) or $x = 1$ (the whole vertical line). Equilibrium: the constant function $y = 0$. Check: left side $(0)' = 0$; right side $0 \cdot (x - 1) = 0$ ✓ — a solution that starts at height 0 stays there forever. **Answer: flat along $y = 0$ and along $x = 1$; the equilibrium solution is $y = 0$.**

10. $dy/dx = -x/y$: (a) zero/undefined slopes, (b) solution with $y(0) = -4$, (c) domain.

Work: (a) slope $-x/y = 0$ where $x = 0$ (with $y \neq 0$); undefined where $y = 0$ (division by zero — vertical tangents on the x -axis). (b) $y \, dy = -x \, dx \rightarrow y^2/2 = -x^2/2 + C \rightarrow x^2 + y^2 = C$; through $(0, -4)$: $C = 16 \rightarrow y = \pm\sqrt{16 - x^2}$; $y(0) = -4 < 0$ picks the MINUS root: $y = -\sqrt{16 - x^2}$. Check: $dy/dx = x/\sqrt{16 - x^2} = -x/(-\sqrt{16 - x^2}) = -x/y$ ✓, $y(0) = -4$ ✓. (c) Need $16 - x^2 > 0$. **Answer: (a) zero on $x = 0$, undefined on $y = 0$; (b) $y = -\sqrt{16 - x^2}$ (bottom half of the circle of radius 4); (c) $-4 < x < 4$.**